

1. CORE INSPECTION METADATA

Timestamp	2026-09-23T22:04:16.219091
Primary Finding	DENT
Severity Layer	HIGH
Detections Logged	1
Analysis Engine	YOLOv11-Nano (Local Edge)

2. ACQUIRED IMAGERY & BOUNDING BOXES



Fig 1. AI-annotated structural imagery overlay.

3. RAW DETECTION COORDINATES

ID	DEFECT CLASS	CONFIDENCE	SEVERITY	BOUNDING BOX [X, Y, W, H]
01	DENT	86.0%	HIGH	[0.430, 0.418, 0.310, 0.207]

4. RAG CONTEXT RETRIEVAL

Query N/A

Source Manual ac_43.13-1b_w-chg1.pdf (Page 150)

Extracted Snippet: *"sustain these stresses, and then allow them to return into the structure. The repair must be equal to the original structure, but not stronger or stiffer, which will cause stress concentrations or alter the resonant frequency of the structure. c. All-metal aircraft are made of very thin sheet metal, and it is possible to restore the strength of the skin without restoring its rigidity. All repairs should be made using the same type and thickness of material that was..."*

5. SYNTHESIZED REPAIR PROTOCOL

Generated via Local Phi-3 Mini strictly grounded in retrieved maintenance documentation.

- 1 Isolate the damaged area**
Clearly mark and isolate the high-severity dent on the aircraft's surface to ensure focused attention during the repair process.
- 2 Remove the dent**
Carefully remove the dented metal using appropriate tools, ensuring not to cause additional damage to the surrounding area.
- 3 Prepare the surface**
Clean and prepare the dented area by removing any debris, rust, or paint, creating a smooth, clean surface for the repair material to adhere to.
- 4 Apply repair material**
Select the same type and thickness of metal sheet used in the original construction and apply it to the dented area, ensuring a seamless integration with the surrounding structure.
- 5 Shape and finish the repair**
Use specialized tools to shape the repair material to match the contours of the original structure, and then finish by polishing or painting to restore the aircraft's appearance.